HashPass: Encryption Base 10

(Original Presentation from November 2024)

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I plan to explain...

Project Description

An overview of my project and how it utilizes hashing

My Code

Brief explanation of code and my thought process while coding

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Project Description

An explanation of my project and how it utilizes hashing

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Password Protection Protocol

In Computer Science, hashing can make the password encryption more secure and trustworthy. This helps prevent the leaking of private information.

My program verifies a user-created password using a small-scale hashing system.

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My Code

Brief explanation of code and my thought process while coding

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Hashing's Helpful Habits

In my code, along with my Decimal to Octal, Hexadecimal, and Binary translations, I included two main functions.

"makePassword" prompts the user to make a password that "checkPassword" will validate the equivalence of using hash-based sums.

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Hashing Utilization *("makePassword" function essentials)*

```
171 // "makePassword" function to make, and verify, a password
172 bool makePassword(string pw, string pw2, int length) {
173     cout << "Enter your password (include 2 or 3 ASCII translations for security): " << endl;
174     cin >> pw;
175     pw2 = pw;
176     length = pw.length();
177
178     cout << endl << "Please re-enter your password: " << endl;
179     cin >> pw;
180     bool pwMatch = checkPassword(pw, pw2, length);
181
182     // Test output: Above statement ("bool pwMatch...")
183     // cout << "pwMatch Bool Value (makePassword function): " << pwMatch << endl << endl;
184
185     if (pwMatch == true) {
186         cout << endl << "Passwords match. Saved successfully." << endl;
187     }
188     else {
189         cout << endl << "Passwords do not match. Try again." << endl;
190     }
191     return pwMatch;
192 }
```

Hashing Utilization (“checkPassword” verification process)

```
// "checkPassword" function to make sure that the hash of two passwords match
bool checkPassword(string pw, string pw2, int n) {
    // Make a hash for both password parameters using ASCII
    int pwHash[100], pw2Hash[100];
    // Bool value to be returned if password hashes match
    bool passMatch = true;
    // Make two sums for comparing the password hashes
    int pwHashSum = 0;
    int pw2HashSum = 0;

    for (int p1 = 0; p1 <= n; p1++) {
        pwHash[p1] = static_cast<int>(pw[p1]) % 10;
    }
    for (int p2 = 0; p2 <= n; p2++) {
        pw2Hash[p2] = static_cast<int>(pw2[p2]) % 10;
    }

    // Call function recursively if password hashes don't match
    for (int a = 0; a < n; a++) {
        pwHashSum = pwHashSum + pwHash[a];
        pw2HashSum = pw2HashSum + pw2Hash[a];
        if (pwHashSum != pw2HashSum) {
            passMatch = false;
            a = n;
        }
    }
    return passMatch;
}
```

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My Results

The code's outputs and my takeaways

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Post-Procedure Processing

Hashing, which is both incredibly fast and effective, has shown me how reliable it is through this project.

Following this slide are sample runs from my code.

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Correct Sample Run *(ASCII and password verification)*

```
Enter a character: ;
(q)uick Mode / (f)unny Mode q
The value given in Octal, base 8, is 73.

The value given in Hexadecimal, base 16, is 3B.

The value given in Binary, base 2, is 111011.

Enter your password (include 2 or 3 ASCII translations for security):
733Bsemi

Please re-enter your password:
733Bsemi

Passwords match. Saved successfully.
```

Incorrect Sample Run *(Password detection)*

```
Enter a character: M
(q)uick Mode / (f)unny Mode q
The value given in Octal, base 8, is 115.

The value given in Hexadecimal, base 16, is 4D.

The value given in Binary, base 2, is 1001101.

Enter your password (include 2 or 3 ASCII translations for security):
115M4Dlol

Please re-enter your password:
115MADlol

Passwords do not match. Try again.
Enter your password (include 2 or 3 ASCII translations for security):
```

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Hashing Utilization *(Funny Mode)*

```
Enter a character: (
(q)uick Mode / (f)unny Mode f
The ASCII value of the given character is 40.
Would you like to see your character in Octal (y / n)?
k
Please answer 'y' or 'n'.
n
Wrong answer. >:(
y
The value given in Octal, base 8, is 50.

Would you like to see your character in Hexadecimal (y / n)?
n
Wrong answer. >:(
y
The value given in Hexadecimal, base 16, is 28.

Would you like to see your character in Binary (y / n)?
n
Wrong answer. >:(
n
Wrong answer. >:(
y
The value given in Binary, base 2, is 101000.

Enter your password (include 2 or 3 ASCII translations for security):
```

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